

Модуль 14: K8s Networking (Ingress) и Helm

Задача (Теория): NodePort vs LoadBalancer vs Ingress.

NodePort

Как работает

- Kubernetes открывает порт **на каждой ноде**
- Обычно диапазон: 30000–32767

Пример

http://NODE_IP:30007

Плюсы

- Просто
- Работает без облака

Минусы

- Некрасивые URL
- Нужно знать IP ноды
- Нет маршрутизации по URL
- Неудобно для продакшена

👉 Используется для тестов.

LoadBalancer

Как работает

- В облаке создаётся внешний балансировщик
- Получает публичный IP

Пример

http://35.204.x.x

Плюсы

- Реальный внешний доступ
- Простота

Минусы

- Требуется облако
- Стоит денег
- Один IP на сервис

👉 Используется в продакшене в облаке.

Ingress

Как работает

- L7-балансировщик (HTTP/HTTPS)
- Один IP → много сервисов
- Маршрутизация по URL и Host

Пример

app.com/api → api-service

app.com/web → web-service

Плюсы

- Красивые URL
- Один IP
- TLS/SSL
- Routing

Минусы

- Нужно Ingress Controller

👉 Лучший вариант для веб-приложений.

Задача (YAML - Ingress): Создать app-ingress.yml (kind: Ingress, apiVersion: networking.k8s.io/v1).

Изменено на / вместо /app, так как приложение там

Задача (Тест Ingress): kubectl apply -f app-ingress.yml. Узнать IP Minikube (minikube ip).

```
ikomarov@Cetus:/opt/basic/module13$ curl 192.168.49.2:30237/  
Hello from Docker!ikomarov@Cetus:/opt/basic/module13$
```

Задача (Ingress (Host-based)): Настроить DNS. Добавить \$(minikube ip) my-app.local в /etc/hosts (или C:\...). Изменить Ingress.yml, добавив host: my-app.local.

```
GNU nano 7.2  
192.168.49.2 my-app.local
```

```
ikomarov@Cetus:/opt/basic/module14$ curl http://my-app.local:30237  
Hello from Docker!ikomarov@Cetus:/opt$ sudo nano /etc/hosts
```

Задача (Теория): Что такое Helm?

Helm = пакетный менеджер Kubernetes

👉 Аналог:

- Linux → apt

- Python → pip
- Node → npm

Helm позволяет установить сложные приложения одной командой.

Например:

- PostgreSQL
- Prometheus
- Grafana
- Redis

Вместо 10 YAML файлов → 1 команда.

Helm использует Charts — шаблоны Kubernetes.

```
ikomarov@Cetus:/opt/basic/module14$ helm install my-pg oci://registry-1.docker.io/bitnamicharts/postgresql
Pulled: registry-1.docker.io/bitnamicharts/postgresql:18.3.0
Digest: sha256:4697965e6d18880297ee50154780bb68edcc066d0ac194061c362f2b28373277
NAME: my-pg
LAST DEPLOYED: Mon Feb 16 21:36:28 2026
NAMESPACE: default
STATUS: deployed
REVISION: 1
TEST SUITE: None
NOTES:
CHART NAME: postgresql
CHART VERSION: 18.3.0
APP VERSION: 18.2.0

WARNING: Since August 28th, 2025, only a limited subset of images/charts are available for free.
  Subscribe to Bitnami Secure Images to receive continued support and security updates.
  More info at https://bitnami.com and https://github.com/bitnami/containers/issues/83267

** Please be patient while the chart is being deployed **

PostgreSQL can be accessed via port 5432 on the following DNS names from within your cluster:

  my-pg-postgresql.default.svc.cluster.local - Read/Write connection

To get the password for "postgres" run:

  export POSTGRES_PASSWORD=$(kubectl get secret --namespace default my-pg-postgresql -o jsonpath="{.data.postgres-password}" | base64 -d)

To connect to your database run the following command:

  kubectl run my-pg-postgresql-client --rm --tty -i --restart='Never' --namespace default --image registry-1.docker.io/bitnami/postgresql:latest --env
  --command -- psql --host my-pg-postgresql -U postgres -d postgres -p 5432

  > NOTE: If you access the container using bash, make sure that you execute "/opt/bitnami/scripts/postgresql/entrypoint.sh /bin/bash" in order to avo

To connect to your database from outside the cluster execute the following commands:

  kubectl port-forward --namespace default svc/my-pg-postgresql 5432:5432 &
  PGPASSWORD="$POSTGRES_PASSWORD" psql --host 127.0.0.1 -U postgres -d postgres -p 5432

WARNING: The configured password will be ignored on new installation in case when previous PostgreSQL release was deleted through the helm command. In t
WARNING: Rolling tag detected (bitnami/postgresql:latest), please note that it is strongly recommended to avoid using rolling tags in a production enviro
+info https://techdocs.broadcom.com/us/en/vmware-tanzu/bitnami-secure-images/bitnami-secure-images/services/bsi-doc/apps-tutorials-understand-rolling-ta
WARNING: Rolling tag detected (bitnami/os-shell:latest), please note that it is strongly recommended to avoid using rolling tags in a production environ
+info https://techdocs.broadcom.com/us/en/vmware-tanzu/bitnami-secure-images/bitnami-secure-images/services/bsi-doc/apps-tutorials-understand-rolling-ta

WARNING: There are "resources" sections in the chart not set. Using "resourcesPreset" is not recommended for production. For production installations, p
  - primary.resources
  - readReplicas.resources
+info https://kubernetes.io/docs/concepts/configuration/manage-resources-containers/
ikomarov@Cetus:/opt/basic/module14$ helm list
NAME      NAMESPACE    REVISION    UPDATED                               STATUS          CHART              APP VERSION
my-pg     default      1           2026-02-16 21:36:28.949482931 +0000 UTC deployed        postgresql-18.3.0  18.2.0
ikomarov@Cetus:/opt/basic/module14$ kubectl get pods
```

```

ikomarov@Cetus:/opt/basic/module13$ kubectl get configmaps | grep postgres
postgres-configmap 2 3d5h
ikomarov@Cetus:/opt/basic/module13$ kubectl delete -f postgres-deployment.yml
deployment.apps "postgres-deployment" deleted
ikomarov@Cetus:/opt/basic/module13$ kubectl delete -f postgres-service.yml
service "postgres-db" deleted
ikomarov@Cetus:/opt/basic/module13$ kubectl delete -f postgres-pvc.yml
persistentvolumeclaim "postgres-pvc" deleted
ikomarov@Cetus:/opt/basic/module13$ kubectl delete -f postgres-secret.yml
secret "postgres-secret" deleted
ikomarov@Cetus:/opt/basic/module13$ kubectl delete -f postgres-configmap.yml
configmap "postgres-configmap" deleted
ikomarov@Cetus:/opt/basic/module13$ kubectl get all | grep postgres
pod/my-pg-postgresql-0 1/1 Running 0 5m8s
service/my-pg-postgresql ClusterIP 10.104.186.25 <none> 5432/TCP 5m8s
service/my-pg-postgresql-hl ClusterIP None <none> 5432/TCP 5m8s
statefulset.apps/my-pg-postgresql 1/1 5m8s
ikomarov@Cetus:/opt/basic/module13$ kubectl get pods
NAME READY STATUS RESTARTS AGE
my-app-deployment-d95c7bbf8-hkn88 1/1 Running 3 (66m ago) 3d5h
my-app-deployment-d95c7bbf8-lzh22 1/1 Running 3 (66m ago) 3d5h
my-app-deployment-d95c7bbf8-mwdn2 1/1 Running 3 (66m ago) 3d5h
my-pg-postgresql-0 1/1 Running 0 5m21s

```